

xtabond — Arellano–Bond linear dynamic panel-data estimation

Syntax	Menu	Description	Options
Remarks and examples	Stored results	Methods and formulas	References
Also see			

Syntax

xtabond *depvar* [*indepvars*] [*if*] [*in*] [, *options*]

<i>options</i>	Description
Model	
<u>noconstant</u>	suppress constant term
<u>diffvars</u> (<i>varlist</i>)	already-differenced exogenous variables
<u>inst</u> (<i>varlist</i>)	additional instrument variables
<u>lags</u> (#)	use # lags of dependent variable as covariates; default is <code>lags(1)</code>
<u>maxldep</u> (#)	maximum lags of dependent variable for use as instruments
<u>maxlags</u> (#)	maximum lags of predetermined and endogenous variables for use as instruments
<u>twostep</u>	compute the two-step estimator instead of the one-step estimator
Predetermined	
<u>pre</u> (<i>varlist</i> [...])	predetermined variables; can be specified more than once
Endogenous	
<u>endogenous</u> (<i>varlist</i> [...])	endogenous variables; can be specified more than once
SE/Robust	
<u>vce</u> (<i>vcetype</i>)	<i>vcetype</i> may be <code>gmm</code> or <code>robust</code>
Reporting	
<u>level</u> (#)	set confidence level; default is <code>level(95)</code>
<u>artests</u> (#)	use # as maximum order for AR tests; default is <code>artests(2)</code>
<i>display_options</i>	control spacing and line width
<u>coeflegend</u>	display legend instead of statistics

A panel variable and a time variable must be specified; use `xtset`; see [XT] `xtset`.
indepvars and all *varlists*, except `pre(varlist[...])` and `endogenous(varlist[...])`, may contain time-series operators; see [U] 11.4.4 Time-series varlists. The specification of *depvar*, however, may not contain time-series operators.
`by`, `statsby`, and `xi` are allowed; see [U] 11.1.10 Prefix commands.
`coeflegend` does not appear in the dialog box.
See [U] 20 Estimation and postestimation commands for more capabilities of estimation commands.

Menu

Statistics > Longitudinal/panel data > Dynamic panel data (DPD) > Arellano-Bond estimation

Description

Linear dynamic panel-data models include p lags of the dependent variable as covariates and contain unobserved panel-level effects, fixed or random. By construction, the unobserved panel-level effects are correlated with the lagged dependent variables, making standard estimators inconsistent. [Arellano and Bond \(1991\)](#) derived a consistent generalized method of moments (GMM) estimator for the parameters of this model; **xtabond** implements this estimator.

This estimator is designed for datasets with many panels and few periods, and it requires that there be no autocorrelation in the idiosyncratic errors. For a related estimator that uses additional moment conditions, but still requires no autocorrelation in the idiosyncratic errors, see [\[XT\] xtldpdsys](#). For estimators that allow for some autocorrelation in the idiosyncratic errors, at the cost of a more complicated syntax, see [\[XT\] xtldpd](#).

Options

Model

noconstant; see [\[R\] estimation options](#).

diffvars(*varlist*) specifies a set of variables that already have been differenced to be included as strictly exogenous covariates.

inst(*varlist*) specifies a set of variables to be used as additional instruments. These instruments are not differenced by **xtabond** before including them in the instrument matrix.

lags(#) sets p , the number of lags of the dependent variable to be included in the model. The default is $p = 1$.

maxlddep(#) sets the maximum number of lags of the dependent variable that can be used as instruments. The default is to use all $T_i - p - 2$ lags.

maxlags(#) sets the maximum number of lags of the predetermined and endogenous variables that can be used as instruments. For predetermined variables, the default is to use all $T_i - p - 1$ lags. For endogenous variables, the default is to use all $T_i - p - 2$ lags.

twostep specifies that the two-step estimator be calculated.

Predetermined

pre(*varlist* [, **lagstruct**(*prelags*, *premaxlags*)]) specifies that a set of predetermined variables be included in the model. Optionally, you may specify that *prelags* lags of the specified variables also be included. The default for *prelags* is 0. Specifying *premaxlags* sets the maximum number of further lags of the predetermined variables that can be used as instruments. The default is to include $T_i - p - 1$ lagged levels as instruments for predetermined variables. You may specify as many sets of predetermined variables as you need within the standard Stata limits on matrix size. Each set of predetermined variables may have its own number of *prelags* and *premaxlags*.

Endogenous

endogenous(*varlist* [, **lagstruct**(*endlags*, *endmaxlags*)]) specifies that a set of endogenous variables be included in the model. Optionally, you may specify that *endlags* lags of the specified variables also be included. The default for *endlags* is 0. Specifying *endmaxlags* sets the maximum number of further lags of the endogenous variables that can be used as instruments. The default is to include $T_i - p - 2$ lagged levels as instruments for endogenous variables. You may specify as many sets of endogenous variables as you need within the standard Stata limits on matrix size. Each set of endogenous variables may have its own number of *endlags* and *endmaxlags*.

SE/Robust

`vce(vcetype)` specifies the type of standard error reported, which includes types that are derived from asymptotic theory and that are robust to some kinds of misspecification; see *Remarks and examples* below.

`vce(gmm)`, the default, uses the conventionally derived variance estimator for generalized method of moments estimation.

`vce(robust)` uses the robust estimator. After one-step estimation, this is the Arellano–Bond robust VCE estimator. After two-step estimation, this is the [Windmeijer \(2005\)](#) WC-robust estimator.

Reporting

`level(#)`; see [\[R\] estimation options](#).

`artests(#)` specifies the maximum order of the autocorrelation test to be calculated. The tests are reported by `estat abond`; see [\[XT\] xtabond postestimation](#). Specifying the order of the highest test at estimation time is more efficient than specifying it to `estat abond`, because `estat abond` must refit the model to obtain the test statistics. The maximum order must be less than or equal to the number of periods in the longest panel. The default is `artests(2)`.

display_options: `vsquish` and `nolstretch`; see [\[R\] estimation options](#).

The following option is available with `xtabond` but is not shown in the dialog box:

`coeflegend`; see [\[R\] estimation options](#).

Remarks and examples

[stata.com](http://www.stata.com)

[Anderson and Hsiao \(1981, 1982\)](#) propose using further lags of the level or the difference of the dependent variable to instrument the lagged dependent variables that are included in a dynamic panel-data model after the panel-level effects have been removed by first-differencing. A version of this estimator can be obtained from `xtivreg` (see [\[XT\] xtivreg](#)). [Arellano and Bond \(1991\)](#) build upon this idea by noting that, in general, there are many more instruments available. Building on [Holtz-Eakin, Newey, and Rosen \(1988\)](#) and using the GMM framework developed by [Hansen \(1982\)](#), they identify how many lags of the dependent variable, the predetermined variables, and the endogenous variables are valid instruments and how to combine these lagged levels with first differences of the strictly exogenous variables into a potentially large instrument matrix. Using this instrument matrix, [Arellano and Bond \(1991\)](#) derive the corresponding one-step and two-step GMM estimators, as well as the robust VCE estimator for the one-step model. They also found that the robust two-step VCE was seriously biased. [Windmeijer \(2005\)](#) worked out a bias-corrected (WC) robust estimator for VCEs of two-step GMM estimators, which is implemented in `xtabond`. The test of autocorrelation of order m and the Sargan test of overidentifying restrictions derived by [Arellano and Bond \(1991\)](#) can be obtained with `estat abond` and `estat sargan`, respectively; see [\[XT\] xtabond postestimation](#).

► Example 1: One-step estimator

[Arellano and Bond \(1991\)](#) apply their new estimators and test statistics to a model of dynamic labor demand that had previously been considered by [Layard and Nickell \(1986\)](#) using data from an unbalanced panel of firms from the United Kingdom. All variables are indexed over the firm i and time t . In this dataset, n_{it} is the log of employment in firm i at time t , w_{it} is the natural log of the real product wage, k_{it} is the natural log of the gross capital stock, and ys_{it} is the natural log of industry output. The model also includes time dummies `yr1980`, `yr1981`, `yr1982`, `yr1983`, and `yr1984`. In table 4 of [Arellano and Bond \(1991\)](#), the authors present the results they obtained from several specifications.

In column a1 of table 4, Arellano and Bond report the coefficients and their standard errors from the robust one-step estimators of a dynamic model of labor demand in which n_{it} is the dependent variable and its first two lags are included as regressors. To clarify some important issues, we will begin with the homoskedastic one-step version of this model and then consider the robust case. Here is the command using xtabond and the subsequent output for the homoskedastic case:

```
. use http://www.stata-press.com/data/r13/abdata
. xtabond n l(0/1).w l(0/2).(k ys) yr1980-yr1984 year, lags(2) noconstant
Arellano-Bond dynamic panel-data estimation      Number of obs      =      611
Group variable: id                               Number of groups     =      140
Time variable: year
Obs per group:      min =      4
                   avg =  4.364286
                   max =      6
Number of instruments =      41                  Wald chi2(16)         =    1757.07
                                                Prob > chi2           =     0.0000
```

One-step results						
n	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
n						
L1.	.6862261	.1486163	4.62	0.000	.3949435	.9775088
L2.	-.0853582	.0444365	-1.92	0.055	-.1724523	.0017358
w						
--.	-.6078208	.0657694	-9.24	0.000	-.7367265	-.4789151
L1.	.3926237	.1092374	3.59	0.000	.1785222	.6067251
k						
--.	.3568456	.0370314	9.64	0.000	.2842653	.4294259
L1.	-.0580012	.0583051	-0.99	0.320	-.172277	.0562747
L2.	-.0199475	.0416274	-0.48	0.632	-.1015357	.0616408
ys						
--.	.6085073	.1345412	4.52	0.000	.3448115	.8722031
L1.	-.7111651	.1844599	-3.86	0.000	-1.0727	-.3496304
L2.	.1057969	.1428568	0.74	0.459	-.1741974	.3857912
yr1980	.0029062	.0212705	0.14	0.891	-.0387832	.0445957
yr1981	-.0404378	.0354707	-1.14	0.254	-.1099591	.0290836
yr1982	-.0652767	.048209	-1.35	0.176	-.1597646	.0292111
yr1983	-.0690928	.0627354	-1.10	0.271	-.1920521	.0538664
yr1984	-.0650302	.0781322	-0.83	0.405	-.2181665	.0881061
year	.0095545	.0142073	0.67	0.501	-.0182912	.0374002

```
Instruments for differenced equation
GMM-type: L(2/.)n
Standard: D.w LD.w D.k LD.k L2D.k D.ys LD.ys L2D.ys D.yr1980
          D.yr1981 D.yr1982 D.yr1983 D.yr1984 D.year
```

The coefficients are identical to those reported in column a1 of table 4, as they should be. Of course, the standard errors are different because we are considering the homoskedastic case. Although the moment conditions use first-differenced errors, xtabond estimates the coefficients of the level model and reports them accordingly.

The footer in the output reports the instruments used. The first line indicates that xtabond used lags from 2 on back to create the GMM-type instruments described in [Arellano and Bond \(1991\)](#) and [Holtz-Eakin, Newey, and Rosen \(1988\)](#); also see [Methods and formulas](#) in [\[XT\] xtldpd](#). The second and third lines indicate that the first difference of all the exogenous variables were used as standard instruments. GMM-type instruments use the lags of a variable to contribute multiple columns to the

instrument matrix, whereas each standard instrument contributes one column to the instrument matrix. The notation `L(2/.) .n` indicates that GMM-type instruments were created using lag 2 of `n` from on back. `(L(2/4) .n` would indicate that GMM-type instruments were created using only lags 2, 3, and 4 of `n`.)

After `xtabond`, `estat sargan` reports the Sargan test of overidentifying restrictions.

```
. estat sargan
Sargan test of overidentifying restrictions
H0: overidentifying restrictions are valid
      chi2(25)      = 65.81806
      Prob > chi2   = 0.0000
```

Only for a homoskedastic error term does the Sargan test have an asymptotic chi-squared distribution. In fact, [Arellano and Bond \(1991\)](#) show that the one-step Sargan test overrejects in the presence of heteroskedasticity. Because its asymptotic distribution is not known under the assumptions of the `vce(robust)` model, `xtabond` does not compute it when `vce(robust)` is specified. The Sargan test, reported by [Arellano and Bond \(1991, table 4, column a1\)](#), comes from the one-step homoskedastic estimator and is the same as the one reported here. The output above presents strong evidence against the null hypothesis that the overidentifying restrictions are valid. Rejecting this null hypothesis implies that we need to reconsider our model or our instruments, unless we attribute the rejection to heteroskedasticity in the data-generating process. Although performing the Sargan test after the two-step estimator is an alternative, [Arellano and Bond \(1991\)](#) found a tendency for this test to underreject in the presence of heteroskedasticity. (See [\[XT\] xtdpd](#) for an [example](#) indicating that this rejection may be due to misspecification.)

By default, `xtabond` calculates the Arellano–Bond test for first- and second-order autocorrelation in the first-differenced errors. (Use `artests()` to compute tests for higher orders.) There are versions of this test for both the homoskedastic and the robust cases, although their values are different. Use `estat abond` to report the test results.

```
. estat abond
Arellano-Bond test for zero autocorrelation in first-differenced errors
```

Order	z	Prob > z
1	-3.9394	0.0001
2	-.54239	0.5876

H0: no autocorrelation

When the idiosyncratic errors are independently and identically distributed (i.i.d.), the first-differenced errors are first-order serially correlated. So, as expected, the output above presents strong evidence against the null hypothesis of zero autocorrelation in the first-differenced errors at order 1. Serial correlation in the first-differenced errors at an order higher than 1 implies that the moment conditions used by `xtabond` are not valid; see [\[XT\] xtdpd](#) for an [example](#) of an alternative estimation method. The output above presents no significant evidence of serial correlation in the first-differenced errors at order 2.

➤ Example 2: A one-step estimator with robust VCE

Consider the output from the one-step robust estimator of the same model:

```
. xtabond n l(0/1).w l(0/2).(k ys) yr1980-yr1984 year, lags(2) vce(robust)
> noconstant

Arellano-Bond dynamic panel-data estimation      Number of obs      =      611
Group variable: id                               Number of groups     =      140
Time variable: year

Obs per group:      min =      4
                   avg =  4.364286
                   max =      6

Number of instruments =      41                  Wald chi2(16)        =    1727.45
                                                Prob > chi2          =      0.0000

One-step results
                                (Std. Err. adjusted for clustering on id)
```

n	Coef.	Robust Std. Err.	z	P> z	[95% Conf. Interval]	
n						
L1.	.6862261	.1445943	4.75	0.000	.4028266	.9696257
L2.	-.0853582	.0560155	-1.52	0.128	-.1951467	.0244302
w						
--.	-.6078208	.1782055	-3.41	0.001	-.9570972	-.2585445
L1.	.3926237	.1679931	2.34	0.019	.0633632	.7218842
k						
--.	.3568456	.0590203	6.05	0.000	.241168	.4725233
L1.	-.0580012	.0731797	-0.79	0.428	-.2014308	.0854284
L2.	-.0199475	.0327126	-0.61	0.542	-.0840631	.0441681
ys						
--.	.6085073	.1725313	3.53	0.000	.2703522	.9466624
L1.	-.7111651	.2317163	-3.07	0.002	-1.165321	-.2570095
L2.	.1057969	.1412021	0.75	0.454	-.1709542	.382548
yr1980	.0029062	.0158028	0.18	0.854	-.0280667	.0338791
yr1981	-.0404378	.0280582	-1.44	0.150	-.0954307	.0145552
yr1982	-.0652767	.0365451	-1.79	0.074	-.1369038	.0063503
yr1983	-.0690928	.047413	-1.46	0.145	-.1620205	.0238348
yr1984	-.0650302	.0576305	-1.13	0.259	-.1779839	.0479235
year	.0095545	.0102896	0.93	0.353	-.0106127	.0297217

```
Instruments for differenced equation
GMM-type: L(2/.)n
Standard: D.w LD.w D.k LD.k L2D.k D.ys LD.ys L2D.ys D.yr1980
          D.yr1981 D.yr1982 D.yr1983 D.yr1984 D.yr1984 D.yr1984
```

The coefficients are the same, but now the standard errors match that reported in [Arellano and Bond \(1991, table 4, column a1\)](#). Most of the robust standard errors are higher than those that assume a homoskedastic error term.

The Sargan statistic cannot be calculated after requesting a robust VCE, but robust tests for serial correlation are available.

```
. estat abond
Arellano-Bond test for zero autocorrelation in first-differenced errors
```

Order	z	Prob > z
1	-3.5996	0.0003
2	-.51603	0.6058

H0: no autocorrelation

The value of the test for second-order autocorrelation matches those reported in [Arellano and Bond \(1991, table 4, column a1\)](#) and presents no evidence of model misspecification.

◀

► Example 3: The Wald model test

xtabond reports the Wald statistic of the null hypothesis that all the coefficients except the constant are zero. Here the null hypothesis is that all the coefficients are zero, because there is no constant in the model. In our [previous example](#), the null hypothesis is soundly rejected. In column a1 of table 4, Arellano and Bond report a chi-squared test of the null hypothesis that all the coefficients are zero, except the time trend and the time dummies. Here is this test in Stata:

```
. test l.n l2.n w l.w k l.k l2.k ys l.ys l2.ys
( 1) L.n = 0
( 2) L2.n = 0
( 3) w = 0
( 4) L.w = 0
( 5) k = 0
( 6) L.k = 0
( 7) L2.k = 0
( 8) ys = 0
( 9) L.ys = 0
(10) L2.ys = 0
      chi2( 10) =   408.29
Prob > chi2 =    0.0000
```

◀

► Example 4: A two-step estimator with Windmeijer bias-corrected robust VCE

The two-step estimator with the Windmeijer bias-corrected robust VCE of the same model produces the following output:

```

. xtabond n l(0/1).w l(0/2).(k ys) yr1980-yr1984 year, lags(2) twostep
> vce(robust) noconstant
Arellano-Bond dynamic panel-data estimation      Number of obs      =      611
Group variable: id                             Number of groups       =      140
Time variable: year

Obs per group:      min =      4
                   avg =  4.364286
                   max =      6

Number of instruments =      41                  Wald chi2(16)          =    1104.72
                                                Prob > chi2            =      0.0000

Two-step results
                                (Std. Err. adjusted for clustering on id)

```

n	Coef.	WC-Robust Std. Err.	z	P> z	[95% Conf. Interval]	
n						
L1.	.6287089	.1934138	3.25	0.001	.2496248	1.007793
L2.	-.0651882	.0450501	-1.45	0.148	-.1534847	.0231084
w						
--.	-.5257597	.1546107	-3.40	0.001	-.828791	-.2227284
L1.	.3112899	.2030006	1.53	0.125	-.086584	.7091638
k						
--.	.2783619	.0728019	3.82	0.000	.1356728	.4210511
L1.	.0140994	.0924575	0.15	0.879	-.167114	.1953129
L2.	-.0402484	.0432745	-0.93	0.352	-.1250649	.0445681
ys						
--.	.5919243	.1730916	3.42	0.001	.252671	.9311776
L1.	-.5659863	.2611008	-2.17	0.030	-1.077734	-.0542381
L2.	.1005433	.1610987	0.62	0.533	-.2152043	.4162908
yr1980	.0006378	.0168042	0.04	0.970	-.0322978	.0335734
yr1981	-.0550044	.0313389	-1.76	0.079	-.1164275	.0064187
yr1982	-.075978	.0419276	-1.81	0.070	-.1581545	.0061986
yr1983	-.0740708	.0528381	-1.40	0.161	-.1776315	.02949
yr1984	-.0906606	.0642615	-1.41	0.158	-.2166108	.0352896
year	.0112155	.0116783	0.96	0.337	-.0116735	.0341045

```

Instruments for differenced equation
GMM-type: L(2/.)n
Standard: D.w LD.w D.k LD.k L2D.k D.ys LD.ys L2D.ys D.yr1980
          D.yr1981 D.yr1982 D.yr1983 D.yr1984 D.year

```

Arellano and Bond recommend against using the two-step nonrobust results for inference on the coefficients because the standard errors tend to be biased downward (see [Arellano and Bond 1991](#) for details). The output above uses the Windmeijer bias-corrected (WC) robust VCE, which [Windmeijer \(2005\)](#) showed to work well. The magnitudes of several of the coefficient estimates have changed, and one even switched its sign.

The test for autocorrelation presents no evidence of model misspecification:

```
. estat abond
Arellano-Bond test for zero autocorrelation in first-differenced errors
```

Order	z	Prob > z
1	-2.1255	0.0335
2	-.35166	0.7251

H0: no autocorrelation



Manuel Arellano (1957–) was born in Elda in Alicante, Spain. He earned degrees in economics from the University of Barcelona and the London School of Economics. After various posts in Oxford and London, he returned to Spain as professor of econometrics at Madrid in 1991. He is a leading expert on panel-data econometrics.

Stephen Roy Bond (1963–) earned degrees in economics from Cambridge and Oxford. Following various posts at Oxford, he now works mainly at the Institute for Fiscal Studies in London. His research interests include company taxation, dividends, and the links between financial markets, corporate control, and investment.

➤ **Example 5: Including an estimator for the constant**

Thus far we have been specifying the `noconstant` option to keep to the standard Arellano–Bond estimator, which uses instruments only for the differenced equation. The constant estimated by `xtabond` is a constant in the level equation, and it is estimated from the level errors. The output below illustrates that including a constant in the model does not affect the other parameter estimates.

```
. xtabond n l(0/1).w l(0/2).(k ys) yr1980-yr1984 year, lags(2) twostep vce(robust)
Arellano-Bond dynamic panel-data estimation   Number of obs       =       611
Group variable: id                           Number of groups      =       140
Time variable: year

Obs per group:   min =         4
                  avg =    4.364286
                  max =         6

Number of instruments =       42                Wald chi2(16)         =    1104.72
                                                Prob > chi2           =     0.0000

Two-step results
                                (Std. Err. adjusted for clustering on id)
```

	n	Coef.	WC-Robust Std. Err.	z	P> z	[95% Conf. Interval]	
n							
	L1.	.6287089	.1934138	3.25	0.001	.2496248	1.007793
	L2.	-.0651882	.0450501	-1.45	0.148	-.1534847	.0231084
w							
	--.	-.5257597	.1546107	-3.40	0.001	-.828791	-.2227284
	L1.	.3112899	.2030006	1.53	0.125	-.086584	.7091638
k							
	--.	.2783619	.0728019	3.82	0.000	.1356728	.4210511
	L1.	.0140994	.0924575	0.15	0.879	-.167114	.1953129
	L2.	-.0402484	.0432745	-0.93	0.352	-.1250649	.0445681
ys							
	--.	.5919243	.1730916	3.42	0.001	.252671	.9311776
	L1.	-.5659863	.2611008	-2.17	0.030	-1.077734	-.0542381
	L2.	.1005433	.1610987	0.62	0.533	-.2152043	.4162908
	yr1980	.0006378	.0168042	0.04	0.970	-.0322978	.0335734
	yr1981	-.0550044	.0313389	-1.76	0.079	-.1164275	.0064187
	yr1982	-.075978	.0419276	-1.81	0.070	-.1581545	.0061986
	yr1983	-.0740708	.0528381	-1.40	0.161	-.1776315	.02949
	yr1984	-.0906606	.0642615	-1.41	0.158	-.2166108	.0352896
	year	.0112155	.0116783	0.96	0.337	-.0116735	.0341045
	_cons	-21.53725	23.23138	-0.93	0.354	-67.06992	23.99542

```
Instruments for differenced equation
GMM-type: L(2/.)n
Standard: D.w LD.w D.k LD.k L2D.k D.ys LD.ys L2D.ys D.yr1980
          D.yr1981 D.yr1982 D.yr1983 D.yr1984 D.yr1984 D.yr1984
Instruments for level equation
Standard: _cons
```

Including the constant does not affect the other parameter estimates because it is identified only by the level errors; see [XT] [xtdpd](#) for details.

➤ Example 6: Including predetermined covariates

Sometimes we cannot assume strict exogeneity. Recall that a variable, x_{it} , is said to be strictly exogenous if $E[x_{it}\epsilon_{is}] = 0$ for all t and s . If $E[x_{it}\epsilon_{is}] \neq 0$ for $s < t$ but $E[x_{it}\epsilon_{is}] = 0$ for all $s \geq t$, the variable is said to be predetermined. Intuitively, if the error term at time t has some feedback on the subsequent realizations of x_{it} , x_{it} is a predetermined variable. Because unforecastable errors today might affect future changes in the real wage and in the capital stock, we might suspect that the log of the real product wage and the log of the gross capital stock are predetermined instead of strictly exogenous. Here we treat w and k as predetermined and use lagged levels as instruments.

```
. xtabond n l(0/1).ys yr1980-yr1984 year, lags(2) twostep pre(w, lag(1,.))
> pre(k, lag(2,.)) noconstant vce(robust)
```

Arellano-Bond dynamic panel-data estimation

Number of obs	=	611
Group variable: id	Number of groups	= 140
Time variable: year		
Obs per group:	min =	4
	avg =	4.364286
	max =	6
Number of instruments =	83	
Wald chi2(15)	=	958.30
Prob > chi2	=	0.0000

Two-step results

(Std. Err. adjusted for clustering on id)

n	Coef.	WC-Robust Std. Err.	z	P> z	[95% Conf. Interval]	
n						
L1.	.8580958	.1265515	6.78	0.000	.6100594	1.106132
L2.	-.081207	.0760703	-1.07	0.286	-.2303022	.0678881
w						
--.	-.6910855	.1387684	-4.98	0.000	-.9630666	-.4191044
L1.	.5961712	.1497338	3.98	0.000	.3026982	.8896441
k						
--.	.4140654	.1382788	2.99	0.003	.1430439	.6850868
L1.	-.1537048	.1220244	-1.26	0.208	-.3928681	.0854586
L2.	-.1025833	.0710886	-1.44	0.149	-.2419143	.0367477
ys						
--.	.6936392	.1728623	4.01	0.000	.3548354	1.032443
L1.	-.8773678	.2183085	-4.02	0.000	-1.305245	-.449491
yr1980	-.0072451	.017163	-0.42	0.673	-.0408839	.0263938
yr1981	-.0609608	.030207	-2.02	0.044	-.1201655	-.0017561
yr1982	-.1130369	.0454826	-2.49	0.013	-.2021812	-.0238926
yr1983	-.1335249	.0600213	-2.22	0.026	-.2511645	-.0158853
yr1984	-.1623177	.0725434	-2.24	0.025	-.3045001	-.0201352
year	.0264501	.0119329	2.22	0.027	.003062	.0498381

Instruments for differenced equation

GMM-type: L(2/.)n L(1/.)L.w L(1/.)L2.k

Standard: D.ys LD.ys D.yr1980 D.yr1981 D.yr1982 D.yr1983 D.yr1984 D.year

The footer informs us that we are now including GMM-type instruments from the first lag of L.w on back and from the first lag of L2.k on back.



□ Technical note

The above example illustrates that `xtabond` understands `pre(w, lag(1, .))` to mean that L.w is a predetermined variable and `pre(k, lag(2, .))` to mean that L2.k is a predetermined variable. This is a stricter definition than the alternative that `pre(w, lag(1, .))` means only that w is predetermined but includes a lag of w in the model and that `pre(k, lag(2, .))` means only that k is predetermined but includes first and second lags of k in the model. If you prefer the weaker definition, `xtabond` still gives you consistent estimates, but it is not using all possible instruments; see [XT] [xtdpd](#) for an [example](#) of how to include all possible instruments.



➤ Example 7: Including endogenous covariates

We might instead suspect that w and k are endogenous in that $E[x_{it}\epsilon_{is}] \neq 0$ for $s \leq t$ but $E[x_{it}\epsilon_{is}] = 0$ for all $s > t$. By this definition, endogenous variables differ from predetermined variables only in that the former allow for correlation between the x_{it} and the ϵ_{it} at time t , whereas the latter do not. Endogenous variables are treated similarly to the lagged dependent variable. Levels of the endogenous variables lagged two or more periods can serve as instruments. In this example, we treat w and k as endogenous variables.

```
. xtabond n l(0/1).ys yr1980-yr1984 year, lags(2) twostep
> endogenous(w, lag(1,.)) endogenous(k, lag(2,.)) noconstant vce(robust)

Arellano-Bond dynamic panel-data estimation   Number of obs       =       611
Group variable: id                           Number of groups      =       140
Time variable: year

Obs per group:   min =         4
                  avg =    4.364286
                  max =         6

Number of instruments =       71              Wald chi2(15)         =    967.61
                                                Prob > chi2           =     0.0000

Two-step results

                               (Std. Err. adjusted for clustering on id)
```

n	Coef.	WC-Robust Std. Err.	z	P> z	[95% Conf. Interval]	
n						
L1.	.6640937	.1278908	5.19	0.000	.4134323	.914755
L2.	-.041283	.081801	-0.50	0.614	-.2016101	.1190441
w						
--.	-.7143942	.13083	-5.46	0.000	-.9708162	-.4579721
L1.	.3644198	.184758	1.97	0.049	.0023008	.7265388
k						
--.	.5028874	.1205419	4.17	0.000	.2666296	.7391452
L1.	-.2160842	.0972855	-2.22	0.026	-.4067603	-.025408
L2.	-.0549654	.0793673	-0.69	0.489	-.2105225	.1005917
ys						
--.	.5989356	.1779731	3.37	0.001	.2501148	.9477564
L1.	-.6770367	.1961166	-3.45	0.001	-1.061418	-.2926553
yr1980	-.0061122	.0155287	-0.39	0.694	-.0365478	.0243235
yr1981	-.04715	.0298348	-1.58	0.114	-.1056252	.0113251
yr1982	-.0817646	.0486049	-1.68	0.093	-.1770285	.0134993
yr1983	-.0939251	.0675804	-1.39	0.165	-.2263802	.0385299
yr1984	-.117228	.0804716	-1.46	0.145	-.2749493	.0404934
year	.0208857	.0103485	2.02	0.044	.0006031	.0411684

```
Instruments for differenced equation
GMM-type: L(2/.)n L(2/.)L.w L(2/.)L2.k
Standard: D.ys LD.ys D.yr1980 D.yr1981 D.yr1982 D.yr1983 D.yr1984
D.year
```

Although some estimated coefficients changed in magnitude, none changed in sign, and these results are similar to those obtained by treating w and k as predetermined. ◀

The Arellano–Bond estimator is for datasets with many panels and few periods. (Technically, the large-sample properties are derived with the number of panels going to infinity and the number of

➤ Example 9: Missing observations in the middle of panels

xtabond handles data in which there are missing observations in the middle of the panels. In this example, we deliberately set the dependent variable to missing in the year 1980:

```
. replace n=. if year==1980
(140 real changes made, 140 to missing)

. xtabond n l(0/1).w l(0/2).(k ys) yr1980-yr1984 year, lags(2) noconstant
> vce(robust)
note: yr1980 dropped from div() because of collinearity
note: yr1981 dropped from div() because of collinearity
note: yr1982 dropped from div() because of collinearity
note: yr1980 dropped because of collinearity
note: yr1981 dropped because of collinearity
note: yr1982 dropped because of collinearity

Arellano-Bond dynamic panel-data estimation Number of obs      =      115
Group variable: id              Number of groups       =      101
Time variable: year

Obs per group:   min =      1
                  avg =  1.138614
                  max =      2

Number of instruments =      18              Wald chi2(12)         =      44.48
                                                Prob > chi2          =      0.0000

One-step results
                                (Std. Err. adjusted for clustering on id)
```

n	Coef.	Robust Std. Err.	z	P> z	[95% Conf. Interval]	
n						
L1.	.1790577	.2204682	0.81	0.417	-.253052	.6111674
L2.	.0214253	.0488476	0.44	0.661	-.0743143	.1171649
w						
--.	-.2513405	.1402114	-1.79	0.073	-.5261498	.0234689
L1.	.1983952	.1445875	1.37	0.170	-.0849912	.4817815
k						
--.	.3983149	.0883352	4.51	0.000	.2251811	.5714488
L1.	-.025125	.0909236	-0.28	0.782	-.203332	.1530821
L2.	-.0359338	.0623382	-0.58	0.564	-.1581144	.0862468
ys						
--.	.3663201	.3824893	0.96	0.338	-.3833451	1.115985
L1.	-.6319976	.4823958	-1.31	0.190	-1.577476	.3134807
L2.	.5318404	.4105269	1.30	0.195	-.2727775	1.336458
yr1983	-.0047543	.024855	-0.19	0.848	-.0534692	.0439606
yr1984	0	(omitted)				
year	.0014465	.010355	0.14	0.889	-.0188489	.0217419

```
Instruments for differenced equation
GMM-type: L(2/.)n
Standard: D.w LD.w D.k LD.k L2D.k D.ys LD.ys L2D.ys D.yr1983
D.yr1984 D.year
```

There are two important aspects to this example. First, xtabond reports that variables have been dropped from the model and from the div() instrument list. For xtabond, the div() instrument list is the list of instruments created from the strictly exogenous variables; see [\[XT\] xtdpd](#) for more about the div() instrument list. Second, because xtabond uses time-series operators in its computations, if statements and missing values are not equivalent. An if statement causes the false observations to

be excluded from the sample, but it computes the time-series operators wherever possible. In contrast, missing data prevent evaluation of the time-series operators that involve missing observations. Thus the example above is not equivalent to the following one:

```
. use http://www.stata-press.com/data/r13/abdata, clear
. xtabond n l(0/1).w l(0/2).(k ys) yr1980-yr1984 year if year!=1980,
> lags(2) noconstant vce(robust)
note: yr1980 dropped from div() because of collinearity
note: yr1980 dropped because of collinearity

Arellano-Bond dynamic panel-data estimation   Number of obs       =       473
Group variable: id                           Number of groups      =       140
Time variable: year

Obs per group:   min =         3
                  avg =    3.378571
                  max =         5

Number of instruments =       37              Wald chi2(15)         =    1041.61
                                                Prob > chi2          =     0.0000
```

One-step results

(Std. Err. adjusted for clustering on id)

n	Coef.	Robust Std. Err.	z	P> z	[95% Conf. Interval]	
n						
L1.	.7210062	.1321214	5.46	0.000	.4620531	.9799593
L2.	-.0960646	.0570547	-1.68	0.092	-.2078898	.0157606
w						
--.	-.6684175	.1739484	-3.84	0.000	-1.00935	-.3274849
L1.	.482322	.1647185	2.93	0.003	.1594797	.8051642
k						
--.	.3802777	.0728546	5.22	0.000	.2374853	.5230701
L1.	-.104598	.088597	-1.18	0.238	-.278245	.069049
L2.	-.0272055	.0379994	-0.72	0.474	-.101683	.0472721
ys						
--.	.4655989	.1864368	2.50	0.013	.1001895	.8310082
L1.	-.8562492	.2187886	-3.91	0.000	-1.285067	-.4274315
L2.	.0896556	.1440035	0.62	0.534	-.192586	.3718972
yr1981	-.0711626	.0205299	-3.47	0.001	-.1114005	-.0309247
yr1982	-.1212749	.0334659	-3.62	0.000	-.1868669	-.0556829
yr1983	-.1470248	.0461714	-3.18	0.001	-.2375191	-.0565305
yr1984	-.1519021	.0543904	-2.79	0.005	-.2585054	-.0452988
year	.0203277	.0108732	1.87	0.062	-.0009833	.0416387

Instruments for differenced equation

GMM-type: L(2/.)n

Standard: D.w LD.w D.k LD.k L2D.k D.ys LD.ys L2D.ys D.yr1981
D.yr1982 D.yr1983 D.yr1984 D.year

The year 1980 is dropped from the sample, but when the value of a variable from 1980 is required because a lag or difference is required, the 1980 value is used.

Stored results

`xtabond` stores the following in `e()`:

Scalars

<code>e(N)</code>	number of observations
<code>e(N_g)</code>	number of groups
<code>e(df_m)</code>	model degrees of freedom
<code>e(g_min)</code>	smallest group size
<code>e(g_avg)</code>	average group size
<code>e(g_max)</code>	largest group size
<code>e(t_min)</code>	minimum time in sample
<code>e(t_max)</code>	maximum time in sample
<code>e(chi2)</code>	χ^2
<code>e(arm#)</code>	test for autocorrelation of order #
<code>e(artests)</code>	number of AR tests computed
<code>e(sig2)</code>	estimate of σ_ϵ^2
<code>e(rss)</code>	sum of squared differenced residuals
<code>e(sargan)</code>	Sargan test statistic
<code>e(rank)</code>	rank of <code>e(V)</code>
<code>e(zrank)</code>	rank of instrument matrix

Macros

<code>e(cmd)</code>	<code>xtabond</code>
<code>e(cmdline)</code>	command as typed
<code>e(depvar)</code>	name of dependent variable
<code>e(twostep)</code>	<code>twostep</code> , if specified
<code>e(ivar)</code>	variable denoting groups
<code>e(tvar)</code>	variable denoting time within groups
<code>e(vce)</code>	<code>vcetype</code> specified in <code>vce()</code>
<code>e(vcetype)</code>	title used to label Std. Err.
<code>e(system)</code>	<code>system</code> , if system estimator
<code>e(hascons)</code>	<code>hascons</code> , if specified
<code>e(transform)</code>	specified transform
<code>e(diffvars)</code>	already differenced variables
<code>e(datasignature)</code>	checksum from <code>datasignature</code>
<code>e(properties)</code>	<code>b V</code>
<code>e(estat_cmd)</code>	program used to implement <code>estat</code>
<code>e(predict)</code>	program used to implement <code>predict</code>
<code>e(marginsok)</code>	predictions allowed by <code>margins</code>

Matrices

<code>e(b)</code>	coefficient vector
<code>e(V)</code>	variance–covariance matrix of the estimators

Functions

<code>e(sample)</code>	marks estimation sample
------------------------	-------------------------

Methods and formulas

A dynamic panel-data model has the form

$$y_{it} = \sum_{j=1}^p \alpha_j y_{i,t-j} + \mathbf{x}_{it} \beta_1 + \mathbf{w}_{it} \beta_2 + \nu_i + \epsilon_{it} \quad i = 1, \dots, N \quad t = 1, \dots, T_i \quad (1)$$

where

the α_j are p parameters to be estimated,
 $\mathbf{x}_{it}$ is a $1 \times k_1$ vector of strictly exogenous covariates,
 β_1 is a $k_1 \times 1$ vector of parameters to be estimated,
 $\mathbf{w}_{it}$ is a $1 \times k_2$ vector of predetermined and endogenous covariates,
 β_2 is a $k_2 \times 1$ vector of parameters to be estimated,
 ν_i are the panel-level effects (which may be correlated with the covariates), and
 ϵ_{it} are i.i.d. over the whole sample with variance σ_ϵ^2 .

The ν_i and the ϵ_{it} are assumed to be independent for each i over all t .

By construction, the lagged dependent variables are correlated with the unobserved panel-level effects, making standard estimators inconsistent. With many panels and few periods, estimators are constructed by first-differencing to remove the panel-level effects and using instruments to form moment conditions.

`xtabond` uses a GMM estimator to estimate $\alpha_1, \dots, \alpha_p, \beta_1$, and β_2 . The moment conditions are formed from the first-differenced errors from (1) and instruments. Lagged levels of the dependent variable, the predetermined variables, and the endogenous variables are used to form GMM-type instruments. See [Arellano and Bond \(1991\)](#) and [Holtz-Eakin, Newey, and Rosen \(1988\)](#) for discussions of GMM-type instruments. First differences of the strictly exogenous variables are used as standard instruments.

`xtabond` uses `xtdpd` to perform its computations, so the formulas are given in [Methods and formulas](#) of [\[XT\] xtdpd](#).

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Also see

[XT] **xtabond postestimation** — Postestimation tools for xtabond

[XT] **xtset** — Declare data to be panel data

[XT] **xtdpd** — Linear dynamic panel-data estimation

[XT] **xtdpdsys** — Arellano–Bover/Blundell–Bond linear dynamic panel-data estimation

[XT] **xtivreg** — Instrumental variables and two-stage least squares for panel-data models

[XT] **xtreg** — Fixed-, between-, and random-effects and population-averaged linear models

[XT] **xtregar** — Fixed- and random-effects linear models with an AR(1) disturbance

[U] **20 Estimation and postestimation commands**